



2003 Network Tariff Report

Produced pursuant to clause 11.3.5 of Volume I of the Office of the Regulator
General's Electricity Distribution Price Determination 2001 – 2005, and
supplementary advisory notes

1 March 2003

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1. Purpose

This report is intended to give guidance to customers, setting out the issues, principles and rationale used by United Energy in setting its Network Use of System (NUOS) tariffs for 2003. Consultation with customers and groups representing customers continued over 2002. However, United Energy welcomes further comments from interested parties and has demonstrated its willingness to incorporate such views into future tariff settings through some of the changes implemented over the past three years.

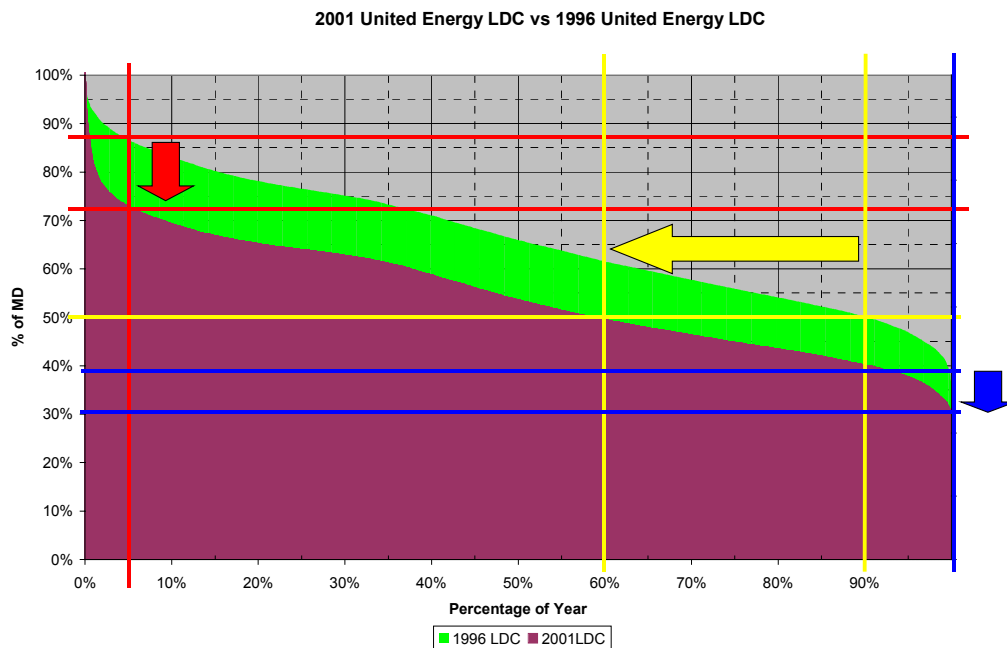
This 2003 tariff report can be read in conjunction with the previous two tariff reports, available on the Essential Services Commission (ESC) (www.esc.vic.gov.au) and United Energy (www.unitedenergy.com.au) websites.

2. Environment review

2003 is the third year of the operation of United Energy's tariff structures since their major restructure in 2001. This section reviews the rationale for those changes.

The 2001 Electricity Price Determination and the consequent price reset applying to Victorian Electricity Distribution businesses provided United Energy an opportunity to review and change its tariff policies and levels. A substantial review of the tariff structures was undertaken in 2000, which culminated in the major tariff restructure implemented in 2001.

One of the main drivers for change was the continuing decline in the network's "load factor". This means that more assets were required to support similar quantities of electricity being transported. This decline was driven by an increase in the amount of cooling load required by United Energy's customers and additional distribution assets have been required to support load only on the hottest summer days. This trend is shown in the following chart.



The above chart shows the Load Duration Curve¹ for the United Energy network comparing 1996 with 2001. Over this 5-year period the Load Factor² of the network fell by 3.8% per year. The red arrow shows the increasing "peakiness" of the load, with the top 5% of the year using 13% of the 1996 Maximum demand compared with 27% in 2001. Further analysis of the numbers behind the chart highlights that nearly 30% of the network was utilised for less than 100 hours (in 2001). In addition, the maximum demand prior to 1997 was recorded in the winter, ever since then it has been recorded in the summer.

¹ The Load Duration Curve shows each half hour period of demand, expressed as a percentage of the capacity of the network, sorted from the largest (100%) to the smallest.

² Load Factor is a measure of the utilisation of the network and can be calculated by dividing the shaded area on the chart by the total area.

The 2001 tariff restructure was carried out against this background and in part was aimed at ensuring that the costs of running the network could be made transparent to all of the users of the network. United Energy wanted to ensure that prices charged for use of the Distribution system better reflected the costs of providing network capacity.

The principles applied in tariff setting policy are further discussed in the next section.

3. Tariff setting principles (basis for tariff policies)

The principles used by United Energy in the tariff setting process can be grouped under the following headings:

- Compliance with the requirements of the Essential Services Commission (ESC);
- Greater customer choice and flexibility;
- Market equity for all customers;
- Reflecting the cost of supplying the network capacity
- Encouraging some behavioural elasticity in the market;
- Recognising the need for practicality; and
- Seeking the most environmentally positive responses

Understandably, not all principles can be satisfied to their full extent, while fully complying with the ESC's requirements, particularly as some pull in opposite directions. United Energy however has sought to achieve as much as possible in satisfying these principles within the given constraints.

UE supports the ESC's pragmatic approach to tariff setting – particularly recognising that there is no 'one best answer' or method. Rather, the ESC has set guidelines for determining an envelope of acceptability using upper and lower bounds, within which tariffs should fall.

In the setting tariff rates, an extensive set of 'rules' specified by the ESC must be complied with. These rules limit the extent to which tariffs can change, particularly any average price increases. Within these constraints, distributors can design further changes that permit innovation and achievement of other objectives.

The design of the tariffs sought to incorporate the following policy principles:

- Regulatory compliance. Compliance with the regulatory requirements of the Office of the Regulator General (ORG) and Essential Services Commission (ESC), as specified in and interpreted from Volumes I & II of the Electricity Distribution Price Determination, and subsequent partial re-determination.

- Customer choice. Where feasible provide customers with meaningful choices of tariff option based upon behavioural response and shared benefits.

- Market equity. Having regard to the previous period price levels and hence relative changes. Further, pricing should apply to all retailers in a neutral manner and not impede the viability of full retail contestability.

- Cost reflectivity. Seek appropriate mechanisms for achieving inter-customer group equity in the allocation of distribution cost-recovery requirements. Pricing to recognise cost-reflectivity, within the constraints imposed by interpretation of regulatory requirements, customer group averaging assumptions, cost allocation methodologies employed and existing conditions, such as existing metering. Have due regard to reasonable economic bounds on upper and lower limits, particularly as pertaining to efficient investment signalling and network usage.

- Behavioural elasticity. Seek to utilise rational consumer behavioural elasticity, both in terms of usage pattern responses to pricing signals and tariff switching.
- Practicality. Seek to simplify mechanisms balancing against economic functionality, while having regard to minimising transaction and pricing administration cost.
- Environmental. Within the limitations of the scope and context of electricity distribution pricing, have regard to opportunities to improve utilisation efficiency and accommodate emerging energy technologies, particularly in respect of greenhouse gas emission reduction.

4. Changes in tariffs, tariff structure and new tariffs

The 2003 Network Use of System (NUOS) tariffs continue to reflect the underlying structure of both the Transmission Use of System (TUOS) and Distribution Use of System (DUOS) charges. That is, the structures of the TUOS and DUOS tariffs are identical and the NUOS rates are the simple addition of DUOS and TUOS. It is likely that this convention will continue, unless there are regulatory requirements to make TUOS tariff components even more cost reflective.

Most of the fundamental restructuring of United Energy's tariffs was undertaken with the 2001 tariff set. The 2002 tariff submission contained some minor restructuring including the replacement of the interruptable tariff introduced in 2001 and two new tariffs for small customers. The bulk of the changes envisaged by the 2001 tariff report have now been carried out. The 2003 tariff set for United Energy contains no new tariffs.

However, there has been one minor change in the tariff structure to address an anomaly created as a result of customers switching retailer part way through a month. This change will also prevent a similar problem occurring when a customer's meter is read over a longer period than one month.

Previously, the Summer Demand Incentive Charge (SDIC) was levied on a calendar month basis, with the Standing Charge and Rolling Demand charge expressed as annual charges. The highest demand for the SDIC period in the calendar month was used to calculate the SDIC quantity for billing purposes. This gave rise to a problem in calculating the charges applicable to each Retailer when a customer changed Retailer part way through a calendar month. One of the Retailers would usually receive a lower SDIC charge resulting from the maximum SDIC quantity occurring in the part of the month covered by the other Retailer. A similar problem occurred for SDIC customers on a 90-day reading cycle that did not occur from the beginning of the calendar month.

As a consequence, the Standing Charges, SDIC and Rolling Demand charges are now expressed on a daily basis. This enables the appropriate apportionment of charges to Retailers, whatever the date a customer switches.

The SDIC quantity will be set at the end of the period of each bill. That is, monthly for monthly read customers and quarterly for quarterly read customers. This creates an incentive for quarterly read customers to change to monthly reading, if they believe the increased cost of reading will be off set by the more frequent resetting of the SDIC quantity.

5. Uptake of Time of Use (TOU) tariffs

In 2001 and 2002, United Energy introduced a number of optional Time of Use (TOU) tariffs for customers for use in conjunction with interval (or smart) meters. At the same time, several tariffs incorporating the historical 11pm to 7am Off Peak time zone, relying on obsolete metering technology, were closed to new connections. New connections requiring off peak network tariffs are now required to install an interval meter. United Energy also implemented a policy change whereby new and replacement single-phase connections would have an interval meter installed instead of two accumulation meters and a time switch. It was intended that such a change in connection characteristics would trigger a change in the underlying network tariff to be billed to the customer's Retailer.

Delays with the ESC's formal position paper on the roll out of interval metering, have meant that United Energy's uptake of TOU tariffs has been substantially lower than forecast at the time of introduction of the 2001 tariffs. A further related issue is the reluctance of Retailers to repackage United Energy's TOU tariffs to smaller customers. To date, no Retailer has publicly made an offer based on the low volume TOU tariff. Informal discussions have revealed a number of operational reasons associated with these decisions, including that most customers currently would need to pay additional fees to have an interval meter installed to take advantage of the new tariffs.

United Energy continues to support the concept of interval meter rollout but there are a number of specific operational issues that need to be resolved prior to such a program commencing. The implementation of a new and replacement interval metering programme, subject to an appropriate business case, will result in an efficient outcome, ultimately delivering lower network prices for all customers. We encourage all interested parties to contribute to the ESC consultation process, which is ongoing at the time of this document going to print.

6. Distribution Use of System (DUOS) tariffs

The regulated price control formula allowed United Energy an average increase in the price control for distribution tariffs in 2003 of 5.9% over the 2002 tariffs. The major components of this increase were “CPI – X” (2.6%), “A” (1.3%) and “S” (2.0%). The term “CPI-X” is the underlying price path for distribution tariffs over the regulatory period (2001-2005). The “A” factor relates to a catch up from the previous regulatory period when United Energy did not collect the entire allowed revenue. The “A” factor will not recur. The “S” factor relates to network reliability and reflects the ongoing improvement in network reliability statistics achieved by United Energy’s management of its network assets.

Within the average DUOS tariff increase, the price control formulae allow individual tariffs to increase by a maximum of 7.7%, although the average increase across all customer classes is limited to 5.9%. This allows United Energy some flexibility as to where the average 5.9% DUOS price control increase is incurred.

As noted in the previous section, uptake of United Energy’s optional “TOU” series of tariffs has been below expectations for a number of reasons. As a reward to customers who have taken up the new tariffs and an incentive to customers who have not yet done so, TOU tariffs received below average increases. This is consistent with United Energy’s strategy of encouraging customer choice through the adoption of interval meters. The optional TOU tariff for small customers received the lowest DUOS increase of any tariff class (2.0%). For the other TOU based tariffs, large customers DUOS rates increased by 2.0% (LV-Large) and 2.9% (HV-Large). The Sub-transmission tariff was increased by the maximum allowed (7.7%) as it is still considered to be under-priced.

Because TOU tariffs received below average increases, it follows that other tariffs received above average increases. Reflecting their increasing contribution to co-incident peak demand due to the growth in cooling load amongst small and medium customers and the inability to distinguish between “dumb” profiles, the open non-TOU small and medium tariffs received DUOS increases averaging 6.9%. This treats them all as if cooling load exists at the premise and customers in these classes without cooling load or who are prepared to respond to TOU price signals are incentivised to switch to one of the optional TOU tariffs available.

All of the closed and obsolete tariffs received the maximum allowed increase in DUOS of 7.7%.

7. Transmission Use of System (TUOS) tariffs

The Grid Fees paid by United Energy do not perfectly reflect the underlying structure of the TUOS tariffs. This is an historical anomaly that has been maintained in order to retain the additive nature of structure of the TUOS and DUOS charges. Restructuring the TUOS tariffs to better reflect the underlying charges they are designed to recover would significantly increase the complexity of the NUOS charges faced by customers. No Victorian DB currently perfectly reflects the underlying structure of the Grid Fees charges by SPI Powernet and VENCORP. United Energy has however restricted the application of TUOS rates to those components of the NUOS which best reflect the underlying Grid Fees, (that is Peak Energy, Summer Demand Incentive Charge and Rolling Demand) and so Off Peak energy does not attract TUOS.

As discussed in last year's tariff report, 2001 had a favourable outcome in the Grid Fees paid by United Energy which were then translated into reduced TUOS tariffs for 2002. However, the 2002 Grid Fees paid by United Energy were significantly higher than those paid in 2001 and forecasts for 2003 show a further increase in the Grid Fees likely to be paid. These factors combine to require a significant increase in TUOS tariffs for 2003, in the order of 54%.

As also discussed in last year's tariff report, the asymmetrical nature of the price control mechanism, whereby reductions in TUOS tariffs are passed through immediately but increases may be capped by the regulator, can lead to significant volatility for customers in TUOS prices year on year. This also exposes Distributors to significant regulatory risk in terms of cashflow and distorts TUOS passthrough pricing signals to the market. A smoothed price control mechanism may have mitigated some of these undesirable outcomes.

Early in October 2002, United Energy and one other distributor, applied to the Essential Services Commission (ESC) to lift the rebalancing constraint on TUOS tariffs. On 18 November 2002, the ESC issued their decision on the proposal to increase the rebalancing control for transmission tariffs. Effectively, the decision limited the TUOS increase for United Energy to 24.3%, although a larger increase was required to recover forecast grid fees in 2003. This decision means the United Energy carries the cashflow cost of over \$10m in unfunded grid fees for 2003 and will more than likely need to apply to the ESC for a further increase in the rebalancing control for the 2004 TUOS tariffs.

As a consequence of the above, all transmission tariff components were increased uniformly by the maximum allowed rate of 24.3%.

8. Network Use of System (NUOS) tariffs

NUOS charges are those passed through to retailers by distributors and seen on some customer invoices (unbundled contract customers in the main) and are the sum of Transmission and Distribution charges. Different customer classes have different contributions to overall Transmission requirements and have different TUOS contributions to their final NUOS charges. The annual changes in NUOS tariff can be seen in the following table.

As can be seen, two of the tariffs marked as closed to new connections are paying above their upper bound or “Stand-alone” cost. The LVL1R and LVL2R rates are historical tariffs for “Large” customers that do not incur a demand charge. These tariffs will continue to increase at the maximum rate until the regulatory price control mechanism allows these tariffs to be collapsed into the appropriate LV Medium tariff category or no customers remain on the tariff. However, the regulatory price control mechanism does not allow for compulsory transfer of customers from these obsolete tariffs.

All of the other tariffs sit between the “Avoidable” and “Stand-alone” costs for their tariff class, most comfortably so.

The cost of supply model has not been updated for this report and the previous bounds have been incremented for CPI movements. Where new tariffs (eg: WET, LVkW-RCACOU) do not have customer history data, the annual tariff movement cannot be calculated and have been omitted.

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Table 1: Changes in annual NUOS tariffs and upper and lower bounds

	2002 Avg NUOS (Exc GST) ⁺	2003 Avg NUOS (Exc GST) ⁺⁺			Lower Bound	Upper Bound
Tariff Group	Average Annual Cost (c/kWh)	Average Annual Cost (c/kWh)	%age Change		"Avoidable Cost"	"Standalone C
LVS1R	5.56	6.06	9.0%	Small	0.6	10.6
LVS2R*	3.74	4.08	9.2%			
LVDed*	1.36	1.46	7.7%			
LVkWTOU	4.48	4.64	3.6%			
LVM1R	7.25	7.88	8.8%	Med	0.6	9.6
LVM2R5D*	5.84	6.34	8.6%			
LVM2R7D*	6.95	7.58	9.2%			
LVUnmetered	3.08	3.36	9.0%		1.9	23.6
LVL1R*	4.91	5.42	10.4%	Large	0.5	5.3
LVL2R*	5.11	5.56	8.8%			
LVkVATOU	2.99	3.13	4.7%			
HVkVATOU	1.72	1.82	6.0%	HV	0.4	3.3
STkVATOU	**	**	**			
* Tariff closed to new connections and customers not already taking supply under this tariff						
** Omitted due to small number of customers in this tariff class						
+ Based on 2002 actual outturns, includes some accrual data						
++ Based on estimated 2003 costs						

9. Future direction of United Energy's tariffs

As discussed in the 2002 tariff report, United Energy is committed to seeking beneficial customer outcomes by providing choice and incentives via innovation in its distribution tariffs. Most of the planned initiatives have now been fully implemented and further changes are not envisaged in the short term, particularly prior to the next regulatory price reset.

There is, however, a watching brief on the following issues which may be taken into account in the next annual tariff setting exercise.

- Encouraging an increased uptake of interval-metering based tariffs such as the current 'TOU' series;
- Encouraging customer demand-side initiatives
- Preserving as far as possible a stable commercial climate for investment in DSM equipment or practices
- The SDIC concept will remain, but the time window may be updated from time to time in order that it remain aligned with the key network peak demands
- Cost-of-supply modelling updated to reflect changes in relative contributions from segments
- Further accentuating the emphasis on peak season (summer), day of week and time of day in order to stimulate DSM response
- Properly integrate the contribution that distribution-connected generator customers should be making to the costs of providing network services that all users share and derive value from.
- Continue to develop arrangements for connecting generators within the network, including a DUOS tariff for Distribution Network Connected Generators.
- Further closure of tariffs based on obsolete metering. For example if interval meters became lower cost than current metering technology, or if interval meter rollout became mandated for new and replacement meters, United Energy would close the remaining obsolete tariffs to new connections.

10. Let us know your views

United Energy continues to encourage customers to discuss ideas on how mutual needs may be better met. Many of the 2003 changes will be familiar to those customers that participated in various workshops, seminars and on-on-one sessions hosted by UE during 2002. It is hoped that this can be further enhanced during the next year.

11. Appendix: 2003 NUOS, DUOS and TUOS tariff rates

See the ESC website at:

<http://www.esc.vic.gov.au/apps/page/user/pdf/NetworkTariffScheduleUnitedEnergy.pdf>

Or see the United Energy website at:

http://www.unitedenergy.com.au/industry/network_tariffs/download/Nuos_2003.pdf

12. Appendix: Glossary of terms

DSM Demand-side management. Refers to the ability of customers that consume a product to change behaviours that alter the overall balance between supply and demand. Examples may include the installation of energy management systems, or deferring consumption at time of peak demand.

ESC Essential Services Commission (Previously Office of the Regulator-General)

Load Duration Curve (LDC) A graphical representation of the utilisation of a network. For the time period under observation, the load on the network is shown, sorted by highest to lowest. The resulting area graphically presents the percentage of the available capacity utilised.

Load Factor Load factor is a measure of how 'steady' a load is. It is the extent to which average usage compares to maximum usage. It is generally much more efficient to supply a high load factor (100%=constant unchanging demand) than a 'peaky' (less than 100%) usage pattern. This is because depending upon the time of the peak, the capacity needs to be there all the time to supply the peak, even though most of the time it may not be fully required.

LVS1R etc Abbreviated Tariff names.

- The 1st two letters refer to the voltage level at which the customer is supplied, LV = low voltage 240V(single phase) or 415V(three phase), HV = high voltage 6.6, 11 or 22kV, ST = sub-transmission 22 and 66kV. HOT = tariff variant based on high ambient temperature threshold
- The 2nd section refers to the type of customer, S = small, M = medium, L = large, no letters generally indicates 'generic' interval metered customer.
- 3rd section indicates type of rate structure, 1R = single energy rate, 2R = peak/off-peak rate, kW = interval metered in kW, kVA = interval metered in kVA.

kW Kilowatts, (1,000 Watts) the ISO unit of power measurement

kWh An I.S.O. unit used for the measurement of energy. It is the product of kW times time (in hours).

kVA kilo-Volt-Ampere, the vector product of voltage times current. It is a measure of the proportion of useful (kW) power delivered vs. the current (Amps) drawn by the load, this is known as 'power factor'. The size of assets required is generally a function of the current. Therefore the closer the power factor is to 1.00, the smaller the asset required to supply the load.

SDIC Summer Demand Incentive Charge